Perfect — now we'll consolidate **everything** into **one clean master roadmap** that is:

✅ Research-grade (GenAI papers)
✅ Quant-grade (probability + stochastic + time series)
✅ Startup-ready (LangChain/RAG/Vector DB/Cloud)
✅ Efficient (blogs + problems + coding only)
❌ No long courses
❌ No lecture binging

This will be your **single source of truth roadmap**.

🎯 FINAL TARGET PROFILE

By the end you should be able to:

GenAI Research

- derive ELBO/attention losses
- reproduce papers
- build new architectures

Quant Foundations

- reason probabilistically
- simulate markets
- price options
- model volatility

GenAI Engineering

- build RAG systems
- use LangChain/LangGraph/LlamaIndex
- fine-tune LLaMA
- deploy cloud apps

If you can do all three → you're **rare and extremely employable**.

🧠 MASTER ROADMAP (Consolidated)

Structure per phase:

Resources → Learn → Practice → Implement → Outcome

🔵 PHASE 1 — PROBABILITY (Core engine of ML + Quant)

🔥 Highest ROI subject in entire roadmap

Resources

Primary:

- Sheldon Ross — *First Course in Probability*

Supplement:

- MIT 6.041 notes (PDF only)
- StatQuest (intuition only)

Avoid:
❌ insurance-heavy sections

Learn

- axioms & conditioning
- Bayes
- expectation/variance
- covariance/correlation
- joint distributions
- transformations + Jacobians
- convolution
- LLN/CLT

- tail bounds

Practice (~200 problems)

From Ross:

- conditioning/Bayes → 30
- expectation/variance → 40
- joint distributions → 40
- transformations/Jacobians → 30
- sums/convolution → 30
- CLT/approximations → 30

Manual Derivations (must do by hand)

- Bayes rule
- E[aX+b], Var(X+Y)
- change-of-variables formula
- convolution formula
- CLT sketch proof

Code Implementations (NumPy only)

- distribution sampler
- CLT simulation
- Monte Carlo π
- bootstrap CI
- Bayesian update demo
- compound Poisson simulation

Outcome

You:

- think probabilistically
- understand VAEs/diffusion math
- handle Monte Carlo

- solve quant probability interviews

🔵 PHASE 2 — LINEAR ALGEBRA FOR ML + FINANCE

Resources

- 3Blue1Brown (intuition only)
- Strang problem sets
- Matrix calculus blog notes

Learn

- projections
- eigen/SVD
- covariance matrices
- matrix calculus
- conditioning

Practice (~70 problems)

- eigen → 20
- SVD → 20
- projections → 10
- matrix calculus → 20

Derivations

- PCA objective
- normal equation
- gradient of x^TAx
- covariance formula

Code

- PCA (eigen + SVD)
- low-rank compression
- portfolio covariance risk
- conditioning instability demo

Outcome

You:

- understand embeddings
- reason about attention
- compute portfolio risk

🔵 PHASE 3 — OPTIMIZATION + ML FIRST PRINCIPLES

Resources

- Boyd convex notes (summary)
- CS229 notes
- Karpathy backprop blog

Learn

- convexity
- gradients
- SGD/Adam/AdamW
- bias–variance

Derivations

- logistic gradient
- softmax gradient
- cross entropy
- ELBO

Implement (NumPy only)

- linear regression
- logistic regression
- Naive Bayes
- kNN
- SGD vs Adam comparison

Experiments

- LR sweeps
- overfitting
- noise robustness

Deliverable

Mini research report with plots

Outcome

You think like a researcher, not a library user

🔵 PHASE 4A — GENAI RESEARCH CORE

Resources (blogs/papers only)

- Lilian Weng
- Karpathy GPT build
- Jay Alammar visuals
- Papers With Code

Learn

- backprop deeply
- transformers math
- VAEs
- GANs
- diffusion
- contrastive learning

Implement (from scratch/light PyTorch)

- MLP
- CNN
- Transformer (GPT)
- VAE
- DDPM

Paper reproduction (≥2 mandatory)

Examples:

- DDPM
- CLIP
- SimCLR
- VAE

Outcome

You can read + reproduce research papers

🔵 PHASE 4B — GENAI ENGINEERING STACK (Startup Skills)

🔥 Industry tools layer

Step 1 — Vector Search Foundations

Resources

- Pinecone blogs
- FAISS docs

Implement

- sentence embeddings
- cosine similarity search
- FAISS index

Outcome

Understand vector DBs internally

Step 2 — RAG Systems

Resources

- LlamaIndex docs
- Pinecone RAG tutorials

Projects

- local RAG (PDF Q&A)
- cloud RAG (Pinecone)

- finance news RAG

Outcome

Production LLM pipelines

Step 3 — LangChain + LangGraph

Implement

- tool-using agent
- multi-step workflow
- retrieval + reasoning chain

Outcome

Agent orchestration skills

Step 4 — LLaMA Fine-tuning

Resources

- HuggingFace PEFT docs
- QLoRA blogs

Projects

- LoRA fine-tune
- quantized inference

Outcome

Cheap customization + deployment

Step 5 — Cloud Deploy

Implement

- FastAPI app
- Dockerize
- deploy on AWS/GCP

Outcome

Live GenAI app

🔵 PHASE 5 — FINANCE MATHEMATICS (Trimmed High-ROI)

Learn

- TVM
- discounting
- continuous compounding
- bonds
- duration
- convexity

Skip:
❌ annuity drills
❌ sinking funds

Practice (~50 problems)

Bond pricing & duration only

Code

- bond pricer

- duration calculator
- convexity hedge demo

Outcome

Fixed-income intuition for quant roles

🔵 PHASE 6 — STOCHASTIC CALCULUS (True Quant Core)

Resources

- Shreve problems
- Brownian motion blogs

Learn

- Brownian motion
- Ito lemma
- SDEs
- risk-neutral pricing

Practice (~60 problems)

Code

- GBM simulation
- Monte Carlo option pricing
- Greeks estimation

Outcome

You can price options like a quant

🔵 PHASE 7 — TIME SERIES + MONTE CARLO

Resources

- QuantStart blog

Implement

- ARIMA
- GARCH
- regime switching
- VaR simulation

Outcome

Modern quant/statistical finance ability

🔵 PHASE 8 — SYSTEMS + C++

Resources

- Effective C++ summaries

Practice

- multithreading
- vectorization
- performance benchmarking

Outcome

Quant dev credibility

🎯 FINAL RESULT

After finishing:

You are simultaneously:

GenAI Researcher

- paper reproduction
- model derivation

GenAI Engineer

- RAG
- LangChain
- LLaMA fine-tuning
- deployment

Quant Foundations

- probability
- stochastic calculus
- time series
- Monte Carlo

This combo is **extremely rare and high value**.